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Understanding and applying coulombic efficiency in lithium metal batteries

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Coulombic efficiency (CE) has been widely used in battery research as a quantifiable indicator for the reversibility of batteries. While CE helps to predict the lifespan of a lithium-ion battery, the prediction is not necessarily accurate in a rechargeable lithium metal battery. Here, we discuss the fundamental definition of CE and unravel its true meaning in lithium-ion batteries and a few representative configurations of lithium metal batteries. Through examining the similarities and differences of CE in lithium-ion batteries and lithium metal batteries, we establish a CE measuring protocol with the aim of developing high-energy long-lasting practical lithium metal batteries. The understanding of CE and the CE protocol are broadly applicable in other rechargeable metal batteries including Zn, Mg and Na batteries.

Batteries that can offer superior performance over the lithium-ion battery (LIB) counterpart, or so called beyond LIB technologies, for various applications such as electrical vehicles and grid energy storage are urgently needed. Among many beyond-LIBs, rechargeable lithium metal batteries (LMBs) with the potential to double the energy density of state-of-art LIBs have been extensively investigated. Recent results have demonstrated promising progress[1] with hundreds of stable cycles achieved in realistic high-energy pouch cells[2].

The vast majority of laboratory research has been conducted with coin cells. While coin cells are useful to offer fundamental insights of chemical and electrochemical processes, performance data generated from coin cell studies are problematic. This is because many of the key parameters in coin cells are often not even in the same magnitude as those acceptable in realistic high-energy batteries[3]. This inconsistency in testing conditions leads to many confusing performance results and greatly decelerates research innovation in meaningful directions. The same problem also arises in sodium, magnesium and zinc batteries, in which unrealitic mass/volume ratios between the anode, cathode and electrolyte in coin cells lead to the same reproducibility issues in realistic batteries.

We have previously explored the minimum requirements of major cell components in building high-energy Li metal pouch cells[1] and proposed coin cell testing conditions for LMBs. However, another important attribute of LMBs, the cycling stability, is always challenging to predict. Coulombic efficiency (CE) is usually used to estimate the cycling life of LIB because CE reflects the loss of Li^+ during each cycle. Although CE has also been frequently linked to explain the Li metal cycle life, it is not very effective to monitor the lifespan of a LMB. Even CE itself measured in different groups varies significantly due to the lack of a standard protocol relevant to the realistic high energy LMB conditions.

In this work, we examine CE and its relation to cycle life in both the LIB and several representative LMBs. The electrochemical principles that govern the applicability of CE as a cell cycling monitor are discussed. A protocol of measuring and applying CE in LMBs is also suggested.

Fundementals of CE and its applicability in LIBs

By definition, CE is the ratio of discharge capacity over charge capacity of a specific electrode in a cell. Because capacity is measured by the total charge flow to/from the electrode and the total capacity of LIBs is usually cathode-limiting, CE can be expressed as the ratio between the amount of Li^+ ions or electrons returning to cathode and the amount of Li^+ or electrons departing from cathode in a full cycle, as indicated in equation (1).

$$\begin{aligned} \mathrm{CE} &= \frac{\text{Discharge capacity}}{\text{Charge capacity}} \\ &= \frac{\text{Total number of } Li^+ \text{ ions back to cathode}}{\text{Total number of } Li^+ \text{ ions departing from cathode}} \\ &= \frac{\text{Total number of } e^- \text{ back to cathode}}{\text{Total number of } e^- \text{ departing from cathode}} \end{aligned} \quad (1)$$

Assuming no side reactions on the electrodes in an ideal cell, the flow of Li^+ or electrons should be completely from reversible electrochemical reactions. CE then equals to 100% according to equation (1). In realistic cells, however, side reactions between electrolytes and electrodes are ubiquitous. The side reactions can be either chemical or electrochemical. In the former case, electrons generated from the parasitic reactions may or may not be collected by the current collectors, depending on the reaction pathways. In the latter case, electron loss and acceptance still take place on the current collectors but are irreversible. In particular, some electrolyte decomposition such as electrolyte polymerization could lead to

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the formation of byproducts that are not electrochemically active[4–6], and thus become irreversible. The electrons released from those irreversible electrochemical reactions can be acquired by the current collector and subsequently counted into the CE calculation, which complicates the meaning of the obtained CE.

In a conventional LIB (Fig. 1a), there is no Li metal. Graphite is dominantly used as the anode. All Li^+ ions are originally stored in the cathode side in LIBs. During charge, Li^+ ions de-intercalate from the cathode (for example, $LiNi_xMn_yCo_zO_2$; NMC, $x + y + z = 1$) lattice and insert into the layered structure of graphite (Fig. 1a). Some Li^+ cations are irreversibly consumed for the solid electrolyte interphase (SEI) formation on graphite surfaces during charge. Thereby, in the subsequent discharge process, fewer Li^+ ions return into the NMC structure compared to the amount of Li^+ ions that leave from the NMC. Accordingly, CE is less than 100%.

If electrolytes such as ethylene carbonate and vinylene carbonate are used in a LIB, they are known to form a stable SEI film against graphite anodes, and the irreversible consumption of Li^+ ions will likely become negligible after the formation cycles, that is, the first few cycles at a very slow rate in order to form stable SEI layers. Furthermore, if the Graphite/NMC cell is operated at 4.25 V or below, after the formation cycles, the electrochemical reactions become quite stable with minimum side reactions. This means that the CE also becomes stable thereafter. The majority of electrons in the CE definition (equation 1) can be attributed to the reversible electrochemical reactions and equals the number of Li^+ ions involved in the redox reaction, which directly determines the reversible capacity of the subsequent cycling. Therefore, the remaining capacity in the cell after certain cycling (Fig.1b) can be effectively calculated by using the conventional method (equation 2)[7–9].

$$\text{Capacity retention} = (\text{CE})^n \tag{2}$$

Where CE refers to coulombic efficiency and n is the cycle number. If 1000 stable cycles with more than 90% capacity retention is desired, the averaged CE would have to be at least 99.99%.

If the charge voltage of the Graphite/NMC battery is raised to beyond 4.25 V, the cathode electrolyte interphase (CEI) and sometimes the reorganization of the cathode lattice also begin to consume Li^+ or alter the number of reversible Li^+ ions in the system. The presence of shuttle reactions will also alter CE and thus complicate the relationship between CE and cell cycle life[10]. CE measurements also necessitate high precision along with temperature and pressure control, if an accurate prediction of long-term cycling is desired[11,12]. To apply equation (2), an optimized electrolyte recipe, stable electrodes and consistent testing conditions are all necessary.

Li metal cells

While equation (2) works reasonably well for LIBs, the capacity retention prediction from CE is questionable for LMBs. We show in Figs 1c and 1d that a higher CE does not necessarily correspond to a more stable cycle life.

In a LMB, Li is not only stored in the cathode but also available from Li metal anode (Fig. 1e). During the first charge, some of the Li^+ ions originated from the cathode will unavoidably participate in side reactions at the anode side, resulting in a Li loss. Upon discharge, although fewer original Li^+ ions return to the cathode, additional 'fresh' Li^+ are stripped from the Li metal 'reservoir' to compensate for the partial loss of the cathode-derived Li^+ ions. Therefore, CE, calculated as the ratio of the amount of Li^+ ions returning to the cathode versus the amount of Li^+ ions departing from cathode in a full cycle, cannot reflect the net Li loss during each cycle. As long as there are plenty of Li^+ sources on the Li anode and a sufficient amount of electrolytes, cell cycling will always be able to continue regardless of the value of CE. This explains why an average CE of only 99.69% still maintains at 94% capacity after 100 cycles (Fig. 1c), although theoretically, capacity retention should decrease to only 74% based on equation (2). The same rules apply to any Li metal half cell in which Li metal is used as the anode (reference electrode) to study electrode materials such as Si for LIBs. Therefore, CE in those half cells has no direct relation to the eventual cycling performance of LIBs consisting of the same electrode material, although the half cell provides sufficient Li source to help to assess the individual electrode material itself.

Of note, CE does not always reflect the qualities of SEI and/or CEI. For example, CE in Fig. 1d is higher than in Fig. 1c in the first 20 cycles. But the cell in Fig. 1d lasts a much shorter time than that in Fig. 1c. Although a higher CE is seen in Fig. 1d, the highly resistant SEI film (and sometimes the CEI film as well) derived from carbonate-based electrolyte (Fig. 1d) noticeably accelerates the cell degradation. How long a Li metal cell stably cycles is therefore determined by the depletion rate of Li metal and electrolyte as well as the build-up rate of cell impedance, whichever comes first[3]. The morphology and surface area of deposited Li also impact the consumption rate of electrolyte and the Li[2], although the origin of uneven Li metal deposition is from mass transport in the electrolyte[13]. It is not surprising that, in some cases, Li dendrites form at a CE of close to 100%, which suddenly short the cell internally without any predictive signs[14]. Another important scenario is that sometimes a 'soft short' happens inside the cell but will not immediately short the cell and impacts CE measured. If a 'soft short' occurs inside a Li/NMC metal cell, the charge curve can be extended to a random length depending on how long the soft short lasts. The cell may still be able to discharge, but the CE will be noticeably lower than those in other cycles. If the soft short happens in Li/Li symmetric cells, it is usually reflected by the 'reduced' over-potential for both Li stripping and plating which is in fact a stable internal electronic connection between two Li electrodes[15]. Therefore, CE does not necessarily represent reversibility of the Li metal cell which is determined by many factors including CE.

Anode-free cells

A Li metal cell without the Li metal anode bears some resemblance to a Li-ion cell in that the Li source is only available from the cathode (Fig. 1f). The copper current collector at the anode side often features rough surfaces with inhomogeneous properties (Fig. 2a), leading to the loss of a large amount of Li during each cycle. Figure 2b suggests that even at a shallow deposition of 0.1 mAh cm^{-2}, Li shows preferential nucleation locations instead of homogenously covering the copper surface. In anode-free cells, the deposited Li anode is completely from cathodes. Charged cathodes are facing a thin but very uneven Li deposited on the anode. Therefore, after the first charge, the ratio of areal capacities between nagetive and positive electrode (referred as N/P ratio) in the cell varies significantly at different locations, which may cause uneven utilization of cathodes and thus disastrous capacity loss in realistic batteries. The first stripping of Li from Cu foil also results in a very uneven surface morphology (Fig. 2c), accelerating the uneven distribution of the electrical field at the anode side and worsening Li deposition in the subsequent cycles.

Figure 2d shows a typical cycling and corresponding CE of an anode-free cell. The areal capacity of NMC used is ~4 mAh cm^{-2}, consistent with all other tests in this work. In this configuration, the averaged CE of Cu/NMC coin cell is 97.79% in 100 cycles. By using equation (2), after 100 cycles, the capacity retention of Cu/NMC cell should be about 10.7%, close to the experimentally measured value of 13.1% after 100 cycles.

In anode-free cells, the SEI accumulation on the anode side can significantly increase the cell impedance and thus strongly affect the cycling. In this regard, this unstable SEI layer formed on the anode side is quite different from the traditional SEI in LIBs and is more preferably called reaction or passivation layers instead. For example, when a carbonate-based electrolyte is used, a very low CE

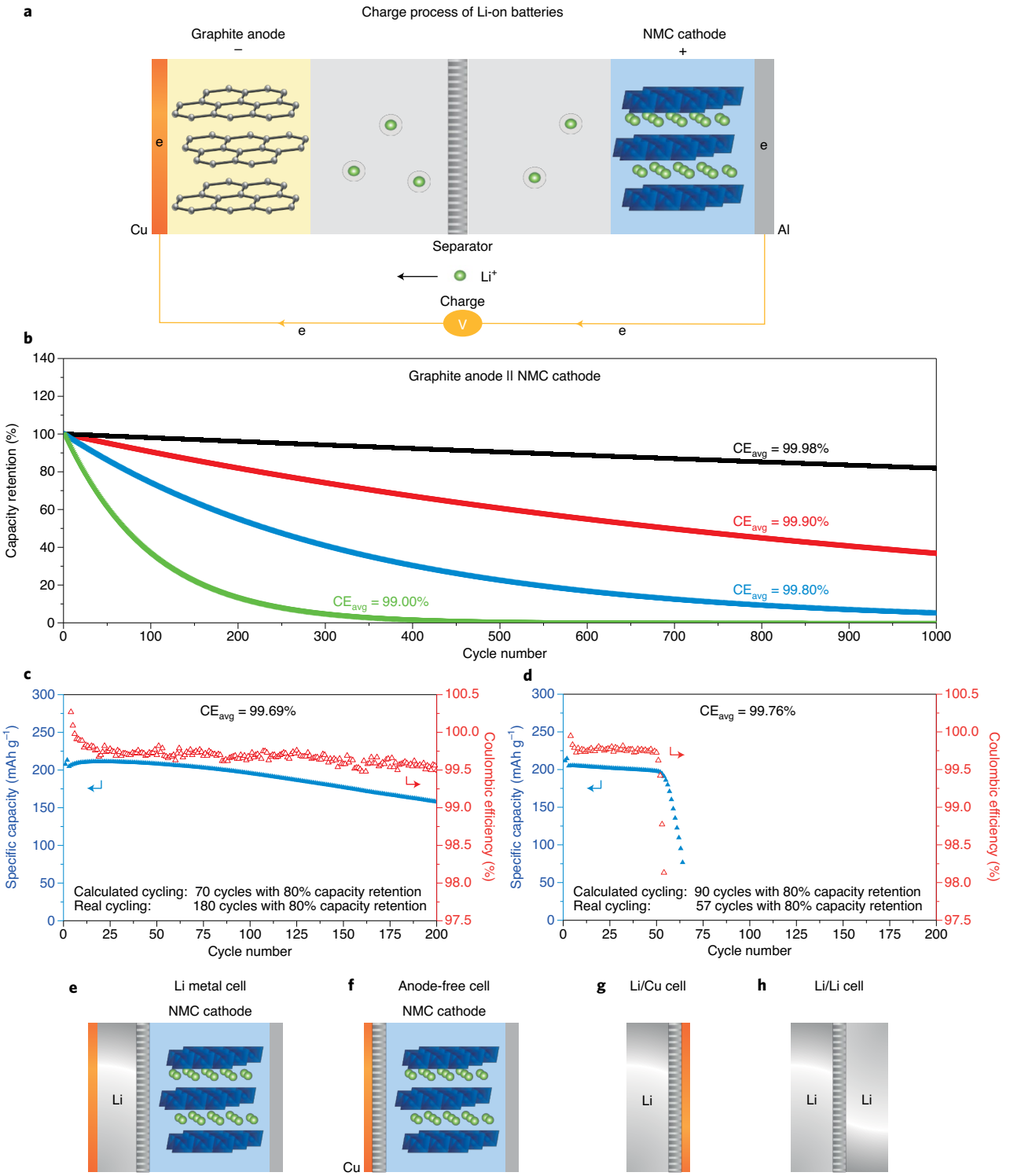

Fig. 1 | Various cell configurations and comparisons between CEs and capacity retentions. a, Charge process of Li-ion batteries (cut-off voltage, 4.25 V). **b**, Cycling of Li-ion batteries with different CE values. Their cycling performances are consistent with prediction from averaged CE based on equation (2). **c**,**d**, Cycling performances and CEs of Li || NMC811 coin cells in two different electrolytes. The cell in **c** has an average CE of 99.69% and cycled stably for more than 180 cycles. The cell in **d** has a much higher average CE of 99.76% but lasts only for 50 cycles. **c** is tested in compatible electrolyte comprised of 1.2 M LiFSI in TEP/BTFE (1:3 by molar ratio), while **d** is tested in conventional carbonate electrolyte comprised of 1.0 M $LiPF_6$ in EC/EMC (3:7 by weight) with 2 wt% VC. The comparison clearly indicates that CE cannot be directly used in Li metal cells to evaluate electrolytes and predict cell cycle life. Both cells after two formation cycles at C/10 were cycled at room temperature by charging at C/10 rate to 4.4 V and at 4.4 V to C/20, and then discharging at C/3 rate to 2.7 V. Li thickness, 50 μm; NMC loading, 4.0 mAh cm^{-2}. **e**–**h**, Different configurations of Li metal cells used in research.

of 92.31% is observed (Fig. 2e) for the limited 40 cycles. Compared to the morphology of Li deposited in a more compatible electrolyte (Fig. 2f), Li deposited in the carbonate-based electrolyte (Fig. 2g) is more porous accommodating more parasitic reactions. Therefore, the insulating passivation layers quickly build up, increase cell impedance and eventually terminate cell operation.

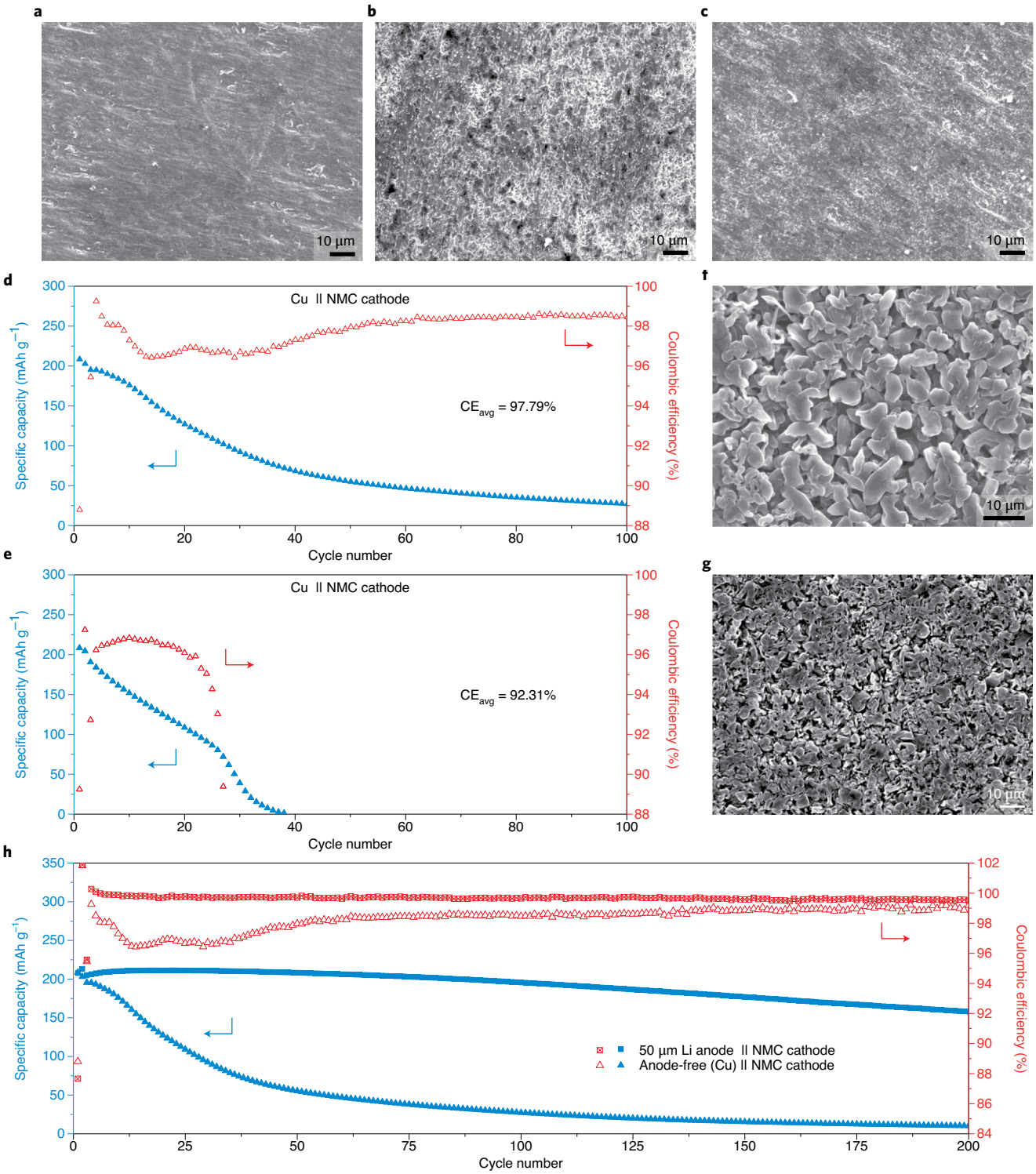

Fig. 2 | Impacts from Cu substrates, electrolytes and Li on measured CE and observed cycling stability of anode-free NMC batteries. a, Morphology of bare Cu. **b**, Li deposited (0.1 mAh cm^{-2}) on Cu in compatible electrolyte comprised of 1.2 M LiFSI in TEP/BTFE (1:3 by molar ratio), which indicates that Li has preferred nucleation sites on Cu. **c**, Remaining Li of **b** after Li being stripped suggesting that 'dead' Li readily formed even after the first cycle. **d**,**e**, Cycling performance and CE of anode-free Cu/NMC811 coin cells in compatible electrolyte: 1.2 M LiFSI in TEP/BTFE (**d**) and conventional carbonate electrolyte: 1.0 M $LiPF_6$ in EC/EMC-2 wt% VC (**e**). **f**,**g**, Morphologies of Li deposited on Cu at depositing capacity of 4.0 mAh cm^{-2} in different electrolytes. Although anode-free cells are used, Li particles formed in compatible electrolyte are still larger than those in carbonate electrolytes thus less amount of side reactions are happening on newly exposed Li surfaces decelerating side reactions during each cycle. **h**, Comparison of cycling performance and CE for 50-μm-Li anode/NMC811 and anode-free Cu/NMC811 coin cells, where the cells after two formation cycles were charged at C/10 to 4.4 V, and then discharged at C/3 to 2.7 V. The addition of 50 μm Li in the anode-free cells significantly boosts the stable cycling and CE.

Another observation in Fig. 2d,e is that CE has no direct relationship with usable capacity. CE can be very high regardless the deliverable capacity is 100% or 50% of its original capacities. The reason is that both the numerator and denominator of equation (1) are changing in each cycle.

Impacts of original Li content in the cell. In 'thin film' cells with 'flooded' electrolytes and 'unlimited' Li metal anode, an artificially long cycling can still be demonstrated even if a 'bad' electrolyte is chosen[3]. When the electrolyte is more than enough, a fast degradation of the anode-free cell is usually caused by fast SEI build-up (cell impedance increase) and/or the quick consumption of Li[16] that converts into 'dead' Li wrapped by thickened SEI layers. If additional Li is provided in the beginning on the anode, CE and cycle life will be readily improved as long as the cell impedance is still not high enough to terminate the cell. For example, the averaged CE in Cu/NMC cell is only 97.79% (Fig. 2h), while that averaged CE of Li/NMC is 99.69% in the same electrolyte. It confirms that an excessive amount of Li in the cell plays a large role in determining the CE and thus cycling stability of a Li metal cell, provided that electrolyte is always sufficient. A recent publication on the quantification of 'dead' Li after cycling[17] provides interesting insights. However, a non-destructive method to quantify residual Li and electrolyte during each cycle without disassembling the cell is still urgently called for.

It is noted that for Li/NMC cell in Fig. 2h, a small capacity increase is displayed for the first ten cycles, while anode-free NMC cell displays continuous capacity fading from the very beginning. The reason is that NMC cathode used here is not only high mass loading (23 mg cm^{-2}) but also dense (porosity: 30%). It is not uncommon to see slight capacity increase in the initial cycles due to the improved electrolyte wetting within the dense cathode. For anode-free NMC cells, however, the discharge capacity is directly related to the amount of Li^+ ions available in the cell which is unfortunately being lost from the very beginning[16,18].

Cathode/electrolyte interface. The choice of cathode also impacts CE of the anode-free cells, although CEI impedance usually does not dominate cell failure as long as the cut-off voltage is lower than 4.3 V[19,20]. Sometimes, a higher cut-off voltage at 4.4 V (versus Li^+/Li) is necessary to extract more energy. Electrolyte decomposition and CEI thickening are therefore unavoidable at high voltages, 'trapping' Li^+ ions on the cathode. After extensive cycling, the impacts from CEI impedance will be displayed by accelerating cell failure. If a low cut-off voltage cathode is used at the very beginning for example, by using $LiFePO_4$ (LFP) cathode, the narrowed electrochemical window (up to 3.8 V) will lead to a higher CE and a longer stable cycling as compared to anode-free NMC cells[21–23].

Li/Cu cells

Unlike in a Li/NMC or anode free cells where CE is calculated based on cathode capacity, CE in a Li/Cu cell (Fig. 1g) is calculated based on the ratio between the amount of stripped Li from the Cu substrate and deposited Li onto the Cu substrate, as described in equation (3). We can also rewrite the equation based on the electrons flowing into or out of the Cu current collector. When plating/stripping Li at low potentials, irreversible electrochemical reactions very often count for an enlarged portion in the denominator and/or the numerator in equation (3). If the amounts of electrons from irreversible electrochemical reactions become substantial, CE measured will not be able to provide quantifiable information of Li^+ ions left in the cell.

$$\mathrm{CE} = \frac{\text{Stripping capacity}}{\text{Depositing capacity}} = \frac{\text{Number of e}^- \text{ from stripping Li}^+ + \text{Number of e}^- \text{ from irreversible electochemical reactions}}{\text{Number of e}^- \text{ from depositing Li}^+ + \text{Number of e}^- \text{ from irreversible electochemical reactions}} \quad (3)$$

A high CE measured in a Li/Cu cell, even with an appropriate electrolyte, cannot guarantee the successful application of the electrolyte in a Li/NMC or anode-free cell. One of the main reasons is that in Li/Cu cells the voltage used for stripping Li is usually less than 1.5 V, avoiding the issues associated with high voltages. However, the CE measured in Li/Cu cells may be very close to those measured in anode-free cells with a LFP cathode given that most electrolytes are stable below 3.8 V[21].

Different protocols and methods have been suggested for Li/Cu cells[24–26], but CE measured, no matter in which way, is not directly relevant to the full cells. It can be used to compare the properties of different electrolytes at low potentials, provided that the testing conditions for example, capacity, rate, electrolyte amount and Li thickness are the same in all tests. For example, the electrolyte amount also significantly impacts the electrochemical properties of a Li/Cu cell. In Fig. 3a, the averaged CE in Li/Cu cell with 75 μL electrolyte is 99.54%, higher than that of the same cell tested with 15 μL of the same electrolyte (CE: 98.60%). The polarization of Li/Cu cells with lean electrolyte also increases more rapidly than those with flooded electrolyte during cycling. The fast depletion of lean electrolyte leads to high polarization, high cell impedance and low CE. Accordingly, cell capacity degrades fast in lean electrolyte, although dense Li is still available. The similar impact of electrolyte content can also be found in Li/NMC cells (Fig. 3b).

It is also important to control the amount of Li deposition in Li/Cu cell testing to obtain meaningful results of cycling. In Li/Cu cell, a fixed amount of Li is deposited on Cu substrate through capacity/time control[24,26]. If the deposited Li on Cu is controlled at 1 mAh cm^{-2} during each cycle while Li source on the Li metal side is as thick as 460 μm (= 92 mAh cm^{-2}), then only 1/92 ≈ 1% of Li anode is being utilized each cycle. As long as there is sufficient Li and electrolyte in the cell, an artificially ultra-long cycling can be achieved. If the goal is to build a high-energy Li metal cell, we recommend that the deposited Li on Cu needs to be comparable with the loading of cathode to be coupled in a full cell, that is, 4 mAh cm^{-2} (ref. [2,3]). The similar criteria need to be considered when evaluating Mg, Zn or Na batteries.

Li/Li cells

Symmetric Li/Li cells (Fig. 1h) are sometimes used to evaluate electrolyte or Li metal properties but it is not quite meaningful. This is because Li/Li cells use capacity/time control for both Li deposition and stripping, so CE is always 100%. However, Li/Li symmetrical cells may be used to assess separator membrane[27] or soild state electrolyte membranes[15] in terms of its capability to induce or mitigate rapid dendrite growth and penetration. But the amount of deposited/stripped Li of each cycle needs to be relevant to the practical applications. Similarly, symmetrical Mg/Mg or Zn/Zn cell does not provide sufficient information to fully evaluate new concepts neither.

Protocol for measuring and applying CE in LMBs

We now propose a general approach of measuring CE in coin cells to assess new concepts of electrodes and electrolytes for high-energy LMBs, as illustrated in Fig. 4. While Li/Li symmetric cells provide limited information, Li/Cu cells are able to obtain some useful information to evaluate the compatibility between the lithium anode and the electrolyte. Note that Li deposition is also happening on the Li side in Li/Cu cells. This means SEI and cell impedance are also being built on Li side upon cycling. A very thick Li foil, coupled with an 'unlimited' amount of electrolyte easily demonstrates a few thousands of artificially stable cycling, which is not recommended. It is suggested that 50-μm-thick Li could be a good start in Li/Cu cells. During each cycle, plating 4 mAh cm^{-2} (≈ 20 μm) of Li on the copper side is a reasonable areal capacity to be coupled with thick cathode for high-energy LMBs[2,3]. The electrolyte does not need

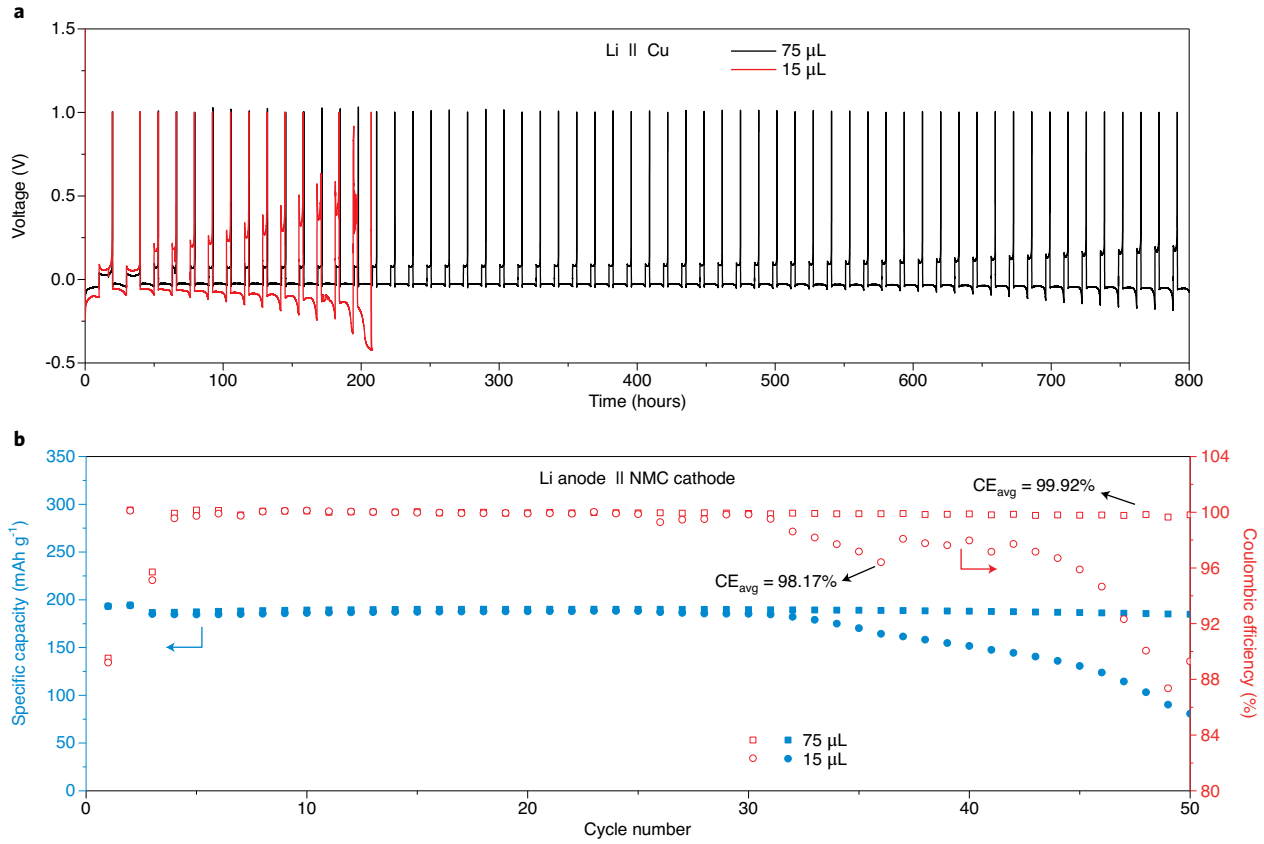

Fig. 3 | Impacts of electrolyte amounts on CE and cycling of Li||Cu and Li||NMC cells. a, Voltage profile of 50 μm Li/Cu coin cells with different amounts of compatible electrolyte comprised of 1.2 M LiFSI in TEP/BTFE (1:3 by molar ratio). Here the Li depositing amount is 4 mAh cm^{-2} and the cut-off voltage of stripping is 1.0 V. After two formation cycles at current density of 0.4 mA cm^{-2}, coin cells were cycled by depositing at current density of 0.4 mA cm^{-2} and stripping at current density of 1.32 mA cm^{-2} up to 1.0 V. The more electrolyte is added in the cell, the lower polarization of each deposition/stripping process and the longer stable cycling can be achieved. **b**, Comparison of cycling performance and CE of 50 μm Li/NMC622 coin cells with different amount of electrolyte. The coin cell with more electrolyte displays higher CE and longer cycling although all the other testing conditions are kept the same. The results indicate that electrolyte content has a considerable impact on CE measurements and cell cycling and thus needs to be consistently controlled in all tests of Li metal batteries.

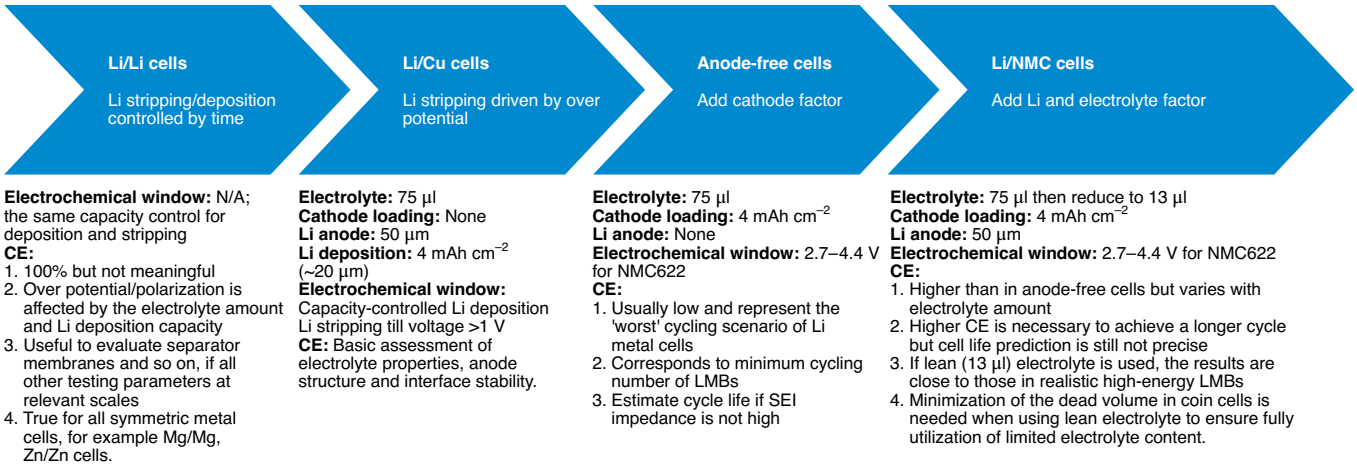

Fig. 4 | A coin cell protocol to measure and interpret CE in LMBs. The suggested coin cell parameters, that is, Li deposition capacity, Li thickness, cathode loading and lean electrolyte amount (13 μl for an electrode area of 1.27 cm^2) are based on the cell design of a realistic high energy LMB of 300 Wh kg^{-1} pouch cell[9,10]. The cathode porosity is controlled at ~35% and separator is 20 μm thick with a porosity of 45%–50%. The electrode area in the coin cell protocol can change but the ratio of lean electrolyte to capacity is ~3 g (Ah)$^{-1}$ or lower. The thickness of Li metal can be thinner than 50 μm but the actual lithium thickness or areal mass should be documented. These suggested parameters may change depending on the cell design, but the cathode loading and lean electrolyte amount will be around the numbers suggested here for designing a high energy LMB.

to be lean at this stage but the amount of electrolyte in all testing should be controlled the same, for example 75 μl which sufficiently 'floods' the entire coin cell.

If a high CE is achievable in Li (50 μm)/Cu cells under strictly controlled conditions, then an anode-free cell can be utilized to understand the 'worst case' scenario in which the minimum cycling number of a LMB can be estimated in realistic conditions. If a LFP is used as the cathode in the anode-free cell, the value of measured CEs will be close to those obtained in Li/Cu cells due to the relatively narrow electrochemical window of LFP. If a NMC cathode is used with a cut-off voltage of > 4.3 V (versus Li^+/Li) being applied, CE would decrease. The cathode loading in anode-free cells is also suggested to be 4 mAh cm^{-2} to be consistent with Li/Cu cells discussed earlier. One can modify the cathode loading based on the specific need but in either Li/Cu or anode-free cells, the amount of deposited Li and electrolyte content need to be consistent and reasonable. High voltage (> 4.3 V) anode-free cell usually cannot last for a long stable cycling but it provides critical information of the minimum cycling stability that a Li metal cell can operate using the combined cathode and electrolyte.

If CE measured in anode-free cells are still high, the success rate of implementing the functioning materials in realistic high-energy LMBs will be high. A Li (50 μm)/NMC (4 mAh cm^{-2}) coin cell can be assembled to validate the new ideas. Because of the increase of Li source on the anode side, the CE and cycle life of Li/NMC cell will increase compared to completely anode-free cell counterpart. Then the amount of the electrolyte in the Li (50 μm)/NMC (4 mAh cm^{-2}) cells can be gradually reduced to become close to the realistic high energy cells. The electrolyte content in a realistic 300 Wh kg^{-1} LMB[2,3] is about 3 g $(Ah)^{-1}$ which translates to about 13 μL in the Li/NMC coin cell assuming electrolyte density is 1.2 g cm^{-3} and the electrode area is 1.27 cm^2[22]. Note that the coin cell parameters in Fig. 4 are only representative values for constructing high-energy Li metal pouch cells. The exact number can vary but should be within a reasonable range[28].

Through this CE measurement protocol, the results obtained from each different cell configuration can be correlated to each other and provide a clear map to understand the range of stable cycling and evaluate its success in realistic batteries. The same principles can be applied to study Mg, Zn and Na batteries where a metal is used as the anode.

Conclusions

The differences and similarities of CE in Li-ion and rechargeable Li metal batteries have been discussed in this work. CE in different cell configurations such as an anode-free cell, Li/Cu cell and Li/Li symmetrical cells are valuable for evaluating materials but the intrinsic principles governing their CE and cycling life are quite different. In addition to electrochemical window and electrolyte compatibility, the original thickness of Li, type and content of electrolyte and cathode mass loading all need to be considered to fully understanding CE and its relation to LMB lifespan. The artificial inflation of CE in Li metal batteries is a general phenomenon in all rechargeable metal batteries which needs to be considered when elaborating testing results. A protocol for measuring CEs in different coin cell configurations has been suggested to unravel the fundamental interrelations of those cells and apply CEs to estimate the range of cycle life for realistic high-energy LMBs. The fundamental understanding of this work is broadly applicable in metal-based battery technologies such as Mg, Zn and Na batteries.

Methods

Cell preparation. Single-side coated electrodes of $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811) and $LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$ (NMC622) (area capacity: 4 mAh cm^{-2}) used as cathode in coin cell were provided by Advanced Battery Facility (ABF) at Pacific Northwest National Laboratory (PNNL). 10 μm Cu foil and 50 μm Li metal foil attached on 10 μm Cu foil were used as anodes in different cell configurations. Polyethylene (PE) film was used as a separator in coin cell. The ingredients of electrolytes such as lithium hexafluorophosphate ($LiPF_6$), ethylene carbonate (EC), ethyl methyl carbonate (EMC), vinlyene carbonate (VC) (BASF Battery Materials), lithium bifluorosulfnyl imde (LiFSI) (Nippon Shokubai), triethylphosphate (TEP) (Sigma Aldrich, ReagentPlus, >99.8%) and bis(2,2,2-trifluoroethyl) ether (BTFE) (SynQuest Labs, 99%) were used as received. The conventional electrolyte with the formula of 1.0 M $LiPF_6$ in EC/EMC (3:7 by weight) + 2 wt% VC and the compatible electrolyte with the formula of 1.2 M LiFSI in TEP/BTFE (a molar ratio of LiFSI:TEP:BTFE at 0.75:1:3) were prepared in argon-filled glovebox (MBraun), where both oxygen and moisture levels were below 0.1 ppm.

The electrochemical performances of all coin cells were evaluated in 2032-type coin cells (National Research Council Canada) on Land battery testers and the cells were filled with 75 μL electrolytes unless otherwise specified (such as 15 μL). 1C rate corresponded to a current density of 4.0 mA cm^{-2}. All Li metal and anode-free/NMC cells, after having two formation cycles at C/10, were cycled by constant current charging at C/10 rate to 4.4 V and then constant voltage charging at 4.4 V to C/20, after that, constant current discharging the cells at C/3 rate to 2.7 V at room temperature in a TestEquity temperature chamber. For the Li/Cu cell, the Li depositing amount on Cu is 4 mAh cm^{-2} and the cut-off voltage of stripping from Cu is 1.0 V. After two formation cycles at current density of 0.4 mA cm^{-2}, the coin cells were cycled by depositing at current density of 0.4 mA cm^{-2} and stripping at current density of 1.32 mA cm^{-2}.

Characterization. For sample preparation, all cycled electrodes were obtained by disassembling the cycled coin cells inside a glovebox and were immediately rinsed with pure EMC to remove the residual electrolyte, and then dried under vacuum. Samples were hermetically sealed inside an air-tight bottle for transferring. Scanning electron microscope (SEM) images were collected on JSM-IT200 (JEOL, Japan) equipped with EDS.

Data availability

All relevant data are included in the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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